VAN NGUYEN NGUYEN

van-nguyen.nguyen@enpc.fr nv-nguyen.github.io (+33)7 69 45 45 75 nv-nguyen nv-nguyen

SUMMARY — I am a PhD student in IMAGINE team of École des Ponts ParisTech (graduation expected: December 2024), advised by Prof. Vincent Lepetit. My research focuses on scalable 6D object detection, 3D reconstruction. During my PhD, I have been fortunate enough to intern twice at Meta Reality Labs: in 2024, where I worked on generic 6D object tracking with Dr. Tomáš Hodaň & Dr. Christian Forster, and in 2022, where I worked on 3D object detection with Dr. Pierre Moulon. I am also co-organizer of BOP challenges (2023, 2024).

RESEARCH EXPERIENCE

Meta Reality Labs — **Jun 2024 – Dec 2024**
Research Intern — *Zurich, Switzerland*

- Subject: Generic 6D object tracking
- Advisors: Dr. Tomáš Hodaň & Dr. Christian Forster

Kyoto University — **Mar 2024 – Apr 2024**
Visiting PhD student, CVLab — *Kyoto, Japan*

- Subject: Articulated object reconstruction
- Advisor: Prof. Ko Nishino

Meta Reality Labs — **May 2022 – Oct 2022**
Research Intern — *Redmond (WA), USA*

- Subject: 3D object detection on Project Aria
- Advisor: Dr. Pierre Moulon

Ecole Polytechnique Fédérale de Lausanne — **Jul 2021 – Aug 2021**
Visiting PhD student, CVLab — *Lausanne, Switzerland*

- Subject: Templates for 3D object pose estimation (CVPR 2022)
- Advisor: Dr. Mathieu Salzmann

Ecole des Ponts ParisTech — **Apr 2020 – Sep 2020**
Research Intern, IMAGINE — *Parisian area, France*

- Subject: 3D scene reconstruction using implicit functions
- Advisors: Dr. Mathieu Aubry & Prof. Vincent Lepetit

Siemens Mobility — **Feb 2019 – Jul 2019**
Research Intern, Optiboard Team — *Parisian area, France*

- Subject: Robust lane detection for self-driving vehicle
- Advisors: Dr. Karim Berkani

Neogia — **Jul 2018 – Sep 2018**
Research Intern, Research and Development Team — *Paris, France*

- Subject: 3D face recontruction from a single image
- Advisors: Dr. Mathieu Bouyrie

EDUCATION

Ecole des Ponts ParisTech — **Oct 2020 – Dec 2024**
Ph.D. in Computer Vision, advised by Prof. Vincent Lepetit — *Parisian area, France*

- Research topics: 6D object pose estimation and tracking, 3D reconstruction

Ecole Normale Supérieure Paris-Saclay — **Sep 2019 – Sep 2020**
Master of Science in Mathematics, Vision, Learning (MVA) — *Parisian area, France*

- Graduated with first-class honors (Mention "Très bien")

INSA Toulouse — **Sep 2014 – Sep 2019**
Master of Engineering in Applied Mathematics — *Toulouse, France**

- Major in Data Science (*Location: 2014-2016 in Vietnam and 2016-2019 in France)

Publications

First-author papers:

GigaPose: Fast and Robust Novel Object Pose Estimation via One Correspondence, CVPR 2024
Van Nguyen Nguyen, Thibault Groueix, Mathieu Salzmann, Vincent Lepetit

NOPE: Novel Object Pose Estimation from a Single Image, CVPR 2024
Van Nguyen Nguyen, Thibault Groueix, Georgy Ponimatkin, Yinlin Hu, Renaud Marlet, Mathieu Salzmann, Vincent Lepetit

CNOS: A Strong Baseline for CAD-based Novel Object Segmentation, ICCVW 2023
Best method for 2D detection/segmentation of novel objects at BOP challenge 2023, R6D Workshop
Van Nguyen Nguyen, Thibault Groueix, Vincent Lepetit, Tomas Hodan

PIZZA: A Powerful Image-only Zero-Shot Zero-CAD Approach to 6DoF Tracking, 3DV 2022 (Oral)
*Van Nguyen Nguyen**, Yuming Du*, Yang Xiao, Michaël Ramamonjisoa, Vincent Lepetit

Templates for 3D Object Pose Estimation Revisited: New Objects and Occlusions, CVPR 2022
Van Nguyen Nguyen, Yinlin Hu, Yang Xiao, Mathieu Salzmann, Vincent Lepetit

Co-author papers:

OpenStreetView-5M: The Many Roads to Global Visual Geolocation, CVPR 2024
G.Astruc, N.Dufour, I.Siglidis, C.Aronssohn, N.Bouia, S.Fu, R.Loiseau, V.N. Nguyen, C.Raude, E. Vincent, L.Xu, H.Zhou, L.Landrieu

BOP Challenge 2023 on Detection, Segmentation and Pose Estimation, CVPRW 2024
T.Hodan, M.Sundermeyer, Y.Labbé, V.N. Nguyen, G. Wang, E.Brachmann, B.Drost, V.Lepetit, C.Rother, J.Matas

Reviewing

- Computer Vision and Pattern Recognition (**CVPR**): 2022, 2023, 2024
- International Conference on Computer Vision (**ICCV**): 2023
- European Conference on Computer Vision (**ECCV**): 2022, 2024
- Neural Information Processing Systems (**NeurIPS**): 2024
- International Conference on Intelligent Robots and Systems (**IROS**): 2024
- Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**): 2022
- International Conference on 3D Vision (**3DV**): 2024
- Asian Conference on Computer Vision (**ACCV**): 2024
- British Machine Vision Conference (**BMVC**): 2022

Other services

- **Teaching**: Image Processing and Artificial Vision (1st Master level) in 2021, 2022
- **Challenges**: Benchmark on 6D Object Pose Estimation, bop.felk.cvut.cz
- **Workshop**: Recovering 6D object pose (ECCV 2024), cmp.felk.cvut.cz/sixd/workshop_2024

Languages

English: Fluent **French:** Fluent **Vietnamese:** Native

References

- **Prof. Vincent Lepetit (ENPC, France)**: vincent.lepetit@enpc.fr
- **Dr. Mathieu Salzmann (EPFL, Switzerland)**: mathieu.salzmann@epfl.ch
- **Dr. Thibault Groueix (Adobe, USA)**: groueix@adobe.com